Paper id: 252561

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM IV) THEORY EXAMINATION 2024-25
MECHANICAL OPERATIONS

TIME: 3 HRS **M.MARKS: 70**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

a.	Define size reduction.	CO1	K1
b.	What is meant by 'shape factor' in particle analysis?	CO2	K2
c.	Define jigging in mineral processing.	CO3	K1
d.	Differentiate between paramagnetic and ferromagnetic materials.	CO4	K1
e.	Name two types of mixers used for high-viscosity fluids	CO5	K2
f.	Mention two differences between crushers and grinders.	CO1	K1
g.	Differentiate between Mixing and Agitation.	CO5	K2

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	Explain the construction, working, and applications of Conical Ball Mill.	CO1	K2
b.	Describe different techniques used for particle size analysis. Compare their advantages and limitations.	CO2	K3
c.	Explain the principles and stages of froth flotation. What are the roles of collectors, frothers, and depressants?	CO3	K3
d.	Derive the Carman-Kozeny and Blake-Plummer equations for pressure drop during fluid flow through a packed bed of spherical particles.	CO4	K3
e.	Explain the design and function of silos, hoppers, and bins. How are they used in industries for bulk solids storage?	CO5	K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Derive and compare the three laws of size reduction: Rittinger's Law, Kick's Law, and Bond's Law. State their assumptions and limitations.	CO1	K3
b.	Discuss the construction, working, and applications of Gyratory Crusher with the help of neat sketch.	CO1	K2

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Derive an expression for screen efficiency and explain the factors affecting it.	CO2	K3
b.	Explain the principle, working, and application of Batch sedimentation.	CO2	K2

5. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain the working principle of electrostatic separators. Discuss various particles charging mechanisms involved in electrostatic separation.	CO3	K3

QP25EP1_311 | 14-Jul-2025 1:30:22 PM | 115.245.67.130

Paper id: 252561

Roll No: | | | | | | | | | | | | |

BTECH
(SEM IV) THEORY EXAMINATION 2024-25
MECHANICAL OPERATIONS

TIME: 3 HRS **M.MARKS: 70**

b.	Describe the construction, working, and applications of wet and dry magnetic separators used in mineral processing.	CO3	K2

6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Explain the derivation of the filtration equation for both constant pressure and constant rate operations.	CO4	K3
b.	A constant pressure filtration test gives data that fit an expression in dt/dv=9.3V+8.5 (t=sec, V=liter) If the resistance of filter medium is assumed unaffected with pressure drop and compressibility coefficient of filter cake is 0.3. Find the time taken for the collection of 3.5 liter of the filtrate at a filtration pressure twice than use in the test.	CO4	K4

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Describe the construction, working, and applications of Ribbon blender.	CO5	K2
b.	Explain the construction, working, and applications of Bucket Elevators with the help of neat sketch.	CO5	K2

QP25EP1_311 | 14-Jul-2025 1:30:22 PM | 115.245.67.130